MODULE 34 · SECTION 5 - MAINTAINING AND AUTOMATING DATA WORKLOADS

Right-Sizing Clusters and Pipelines

Matching cost to the business need: Spark clusters and Spot VMs, Dataflow levers, Managed Airflow environments, storage levers, and seeing where the money goes.

SECTION 5 of 5 EXAM WEIGHT 18% MODULE 34 of 45

admin@sthithapragna.com

Pay for the deadline, not for the peak. **Idle capacity is the most expensive kind.**

What does the deadline really need?

THE BUSINESS NEED	COST LEVER
Done **by morning**, start time flexible	FlexRS, Spot workers, batch priority
Runs **once a night**	A job cluster or serverless, deleted after
Steady all year	Committed use discounts
Duplicates are tolerable	At-least-once streaming
Bursty, unpredictable	Autoscaling with a sensible maximum

THE PATTERN

Read the deadline in the question first. It usually rules out half the options.

A cluster that lives, or a cluster per job?

PERSISTENT CLUSTER

Always on

- Paid every hour, busy or idle
- Shared by many teams and jobs
- Data in HDFS ties you to it
- Suits steady, shared, interactive use

JOB-SCOPED CLUSTER

Create, run, delete

- Paid only while the job runs
- Sized for one job's needs
- Data in Cloud Storage, outlives it
- Or serverless Spark: no cluster at all

REMEMBER

Data in Cloud Storage is what makes deletion safe. Module 15 showed why.

Cheaper workers, with care

- **Spot VMs** as secondary workers cost far less, and can be reclaimed at any time.
- Secondary workers hold no HDFS. Keep preemptible ones under **half** of all workers.
- **Autoscaling** adds and removes workers, but never shrinks an idle cluster to zero.
- **Scheduled deletion**: delete after idling, at a set time, or at a maximum age.

> THE TRAP
>
> Spot workers suit retryable batch work. Never put a hard deadline on them alone.

Larkspur's nightly settlement, right-sized

```
gcloud dataproc clusters create lk-settle-nightly \
    --region=europe-west2 --num-workers=4 \
    --num-secondary-workers=6 --secondary-worker-type=spot \
    --autoscaling-policy=settle-policy \
    --max-idle=15m --max-age=4h \
    --labels=team=settlement,env=prod,workload=nightly
```

3 Six of ten workers are Spot: cheap, and retryable.

5-6 Deleted when idle or too old. Labels show up in billing.

FOR LARKSPUR

Four standard workers carry the deadline. The Spot workers only make it cheaper and faster.

Dataflow cost levers

- **FlexRS** for batch that can wait: a discounted, delayed start, within six hours.
- **At-least-once** streaming, when the sink can absorb duplicates.
- The smallest **machine type** that works, and a cap on **max workers**.
- Run in the **same region** as sources, sinks and the temp bucket.

> THE TRAP
>
> Many tiny batch jobs each pay to start workers. Fewer, larger runs cost less.

Seeing a job's cost

- The **Cost** page in the Dataflow monitoring UI shows each job's estimated cost.
- **Alerts** fire when a job's cost passes a threshold you set.
- **Pilot runs** on a slice of data estimate cost before the full run.
- **Committed use discounts** suit a predictable streaming minimum for a year or more.

> REMEMBER
>
> Data sampling, not logging every element, is the cheap way to debug a pipeline.

Right-sizing an Airflow environment

- Pick the **size preset** from DAG count and peak concurrent tasks.
- **Workers autoscale** between a minimum and a maximum. Keep the minimum low.
- Start with **two schedulers**. In Gen 3, DAG processors do the parsing.
- No pause button: schedulers, web server and database bill while it **exists**.

> THE TIEBREAKER
>
> Airflow should orchestrate, not compute. Heavy work belongs in BigQuery, Dataflow or Spark.

Storage levers, from section three

LEVER	SAVES ON	MODULE
Lifecycle rules and Autoclass	Cloud Storage classes	22
Soft delete off for temporary data	Deleted bytes still billed	22
Partition expiration, no rewrites	BigQuery active storage	23
Physical storage billing	Compressible, rarely changed data	23
Compact small files	Listing time and operations	25

THE PATTERN

Storage costs grow quietly every month. Set the levers once, when the data first lands.

Budgets or billing export?

BUDGETS AND ALERTS

Warn early

- On actual or forecast spend
- Default alerts at 50, 90 and 100%
- Email, or Pub/Sub for automation
- Do not cap spending by themselves

BILLING EXPORT

Explain the bill

- Cost data in a BigQuery dataset
- Labels, projects and services
- Detailed export adds each resource
- Enable early: not fully backfilled

THE TRAP

'Stop spending automatically' needs a Pub/Sub action, or a hard cap such as a reservation maximum.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
Cluster **idle** most of the day	Job-scoped cluster, or scheduled deletion
Batch can **start any time** tonight	FlexRS, or Spot secondary workers
Streaming, **duplicates acceptable**	At-least-once mode
Airflow bill high, **little runs** on it	Smaller preset, low worker minimum
Which **team** spends what?	Labels plus billing export
Warn before the month overruns	Budget with forecast alerts

THE PATTERN

Idle, flexible, tolerant or invisible: each word points at a different lever.

> Larkspur's 20-node Spark cluster runs all day, but the settlement jobs on it take three hours each night.
>
> The input and output data live in Cloud Storage, and the cluster sits idle the rest of the day.

What should the team change?

A. Add an autoscaling policy, so the idle cluster shrinks to zero

B. Switch every worker on the cluster to Spot VMs

C. Create a job cluster each night, with scheduled deletion

D. Buy committed use discounts for all twenty nodes

◆ PAUSE AND DECIDE

What should the team change?

A Autoscaling to zero

Autoscaling never shrinks a cluster to nothing.

B All Spot

Cheaper per hour, still idle 21 hours a day, and fragile.

C Nightly job cluster

Pays for three hours, not twenty four.

D Commit to 20 nodes

Locks in paying for the idle hours.

THE SIGNAL

Idle most of the day, data in Cloud Storage: create per job, delete after.

> A nightly Dataflow batch job rebuilds the tariff analysis. It takes 40 minutes, starts at 22:00, and is only needed by 06:00.
>
> Finance wants the job to cost less, with no change to the code.

What should the team do?

A. Switch the job to at-least-once mode to halve the cost

B. Raise the maximum number of workers, so it finishes sooner

C. Move the job to a region with spare capacity at night

D. Run it with FlexRS, using the cost-optimised goal

◆ PAUSE AND DECIDE

What should the team do?

A At-least-once

A streaming mode, not for batch jobs.

B More workers

Faster, but not cheaper.

C Another region

Moves data across regions, adding cost and risk.

D FlexRS

A discounted rate, starting within six hours.

THE SIGNAL

Batch that can wait hours, cheaper with no code change: FlexRS.

> Larkspur's Managed Airflow environment uses the Large preset with a minimum of 10 workers.
>
> It runs 40 DAGs that mostly trigger BigQuery jobs, and rarely more than a dozen tasks at once.

What should the team change?

A Delete the environment every night, then recreate it

B Move to a smaller preset, and lower the minimum worker count

C Add a third scheduler, so that the tasks finish more quickly

D Run the BigQuery transformations on Airflow workers instead

◆ PAUSE AND DECIDE

What should the team change?

A Delete nightly

Breaks schedules and history, and adds risk.

B Smaller preset, low minimum

Fits forty DAGs, and idle workers stop billing.

C Third scheduler

More fixed cost, and scheduling is not the bottleneck.

D Compute on workers

Moves work to Airflow, raising cost.

THE SIGNAL

An oversized Airflow environment: smaller preset, low worker minimum.

> The finance director wants to know what the settlement team spends each month across Spark, Dataflow and BigQuery,
>
> and to be warned before the month's spend runs over its forecast.

What should the team set up?

A Team labels, billing export, and a budget with forecast alerts

B A separate billing account for each team, checked by hand monthly

C Read the Cost page in the Dataflow monitoring UI once a week

D A single budget on the whole billing account, and nothing else

◆ PAUSE AND DECIDE

What should the team set up?

A Labels, export, budget

Cost by team in SQL, plus early warning.

B Billing account per team

Heavy to run, and still no early warning.

C Dataflow Cost page

Covers Dataflow only, and warns nobody.

D Budget alone

Warns about the total, but cannot split it by team.

THE SIGNAL

Spend by team, plus early warning: labels, billing export and a budget.

Module 34 in five lines

- Pay for the **deadline**, not the peak. Idle capacity costs most.
- Spark: **job clusters** or serverless, Spot secondaries, scheduled deletion.
- Dataflow: **FlexRS** for batch that waits, at-least-once, capped workers.
- Airflow: right **preset**, low worker minimum, no heavy compute on workers.
- **Labels** plus billing export explain the bill. Budgets warn, but do not cap.

COMING UP

Next: module 35, orchestration with Managed Airflow, Workflows and scheduling.